# THE PULLBACK OF A FIBRED CATEGORY ALONG A BASE FUNCTOR IS FIBRED

THE STACKS PROJECT AUTHORS

*Editorial note.* No omission markers in-span; proof complete as written.

*Allowed standard prerequisites (editorial).*

- Strongly cartesian morphisms. categories-definition-cartesian-over-C, categories-section-fibred-categories (cross-chapter, named). Fix the strong cartesian condition and the uniqueness of strongly cartesian lifts used in the converse.
- Composition of strongly cartesian morphisms. categories-lemma-composition-cartesian (cross-chapter, named). Concludes that beta = beta' o iota is strongly cartesian in the converse direction.

*Source reference.* Original reference 02XK: focused-seeds/geometry/S9-stk-fibred-category-pushforward/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 02XM: focused-seeds/geometry/S9-stk-fibred-category-pushforward/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 02XL: focused-seeds/geometry/S9-stk-fibred-category-pushforward/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 02XJ: focused-seeds/geometry/S9-stk-fibred-category-pushforward/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *In the situation above, if $\mathcal{S}$ is a fibred category over $\mathcal{D}$ then $u^p\mathcal{S}$ is a fibred category over $\mathcal{C}$.*

*Proof.* Please take a look at the discussion surrounding Categories, Definitions 02XK and 02XM before reading this proof. Let $(a, \beta) : (U, y) \to (U', y')$ be a morphism of $u^p\mathcal{S}$. We claim that $(a, \beta)$ is strongly cartesian if and only if $\beta$ is

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strongly cartesian. First, assume $\beta$ is strongly cartesian. Consider any second morphism $(a_1, \beta_1) : (U_1, y_1) \to (U', y')$ of $u^p\mathcal{S}$. Then

$$\begin{aligned}
&\mathrm{Mor}_{u^p\mathcal{S}}((U_1, y_1), (U, y)) \\
&= \mathrm{Mor}_{\mathcal{C}}(U_1, U) \times_{\mathrm{Mor}_{\mathcal{D}}(u(U_1), u(U))} \mathrm{Mor}_{\mathcal{S}}(y_1, y) \\
&= \mathrm{Mor}_{\mathcal{C}}(U_1, U) \times_{\mathrm{Mor}_{\mathcal{D}}(u(U_1), u(U))} \mathrm{Mor}_{\mathcal{S}}(y_1, y') \times_{\mathrm{Mor}_{\mathcal{D}}(u(U_1), u(U'))} \mathrm{Mor}_{\mathcal{D}}(u(U_1), u(U)) \\
&= \mathrm{Mor}_{\mathcal{S}}(y_1, y') \times_{\mathrm{Mor}_{\mathcal{D}}(u(U_1), u(U'))} \mathrm{Mor}_{\mathcal{C}}(U_1, U) \\
&= \mathrm{Mor}_{u^p\mathcal{S}}((U_1, y_1), (U', y')) \times_{\mathrm{Mor}_{\mathcal{C}}(U_1, U')} \mathrm{Mor}_{\mathcal{C}}(U_1, U)
\end{aligned}$$

the second equality as $\beta$ is strongly cartesian. Hence we see that indeed $(a, \beta)$ is strongly cartesian. Conversely, suppose that $(a, \beta)$ is strongly cartesian. Choose a strongly cartesian morphism $\beta' : y'' \to y'$ in $\mathcal{S}$ with $p(\beta') = u(a)$. Then bot $(a, \beta) : (U, y) \to (U, y')$ and $(a, \beta') : (U, y'') \to (U, y)$ are strongly cartesian and lift $a$. Hence, by the uniqueness of strongly cartesian morphisms (see discussion in Categories, Section 02XJ) there exists an isomorphism $\iota : y \to y''$ in $\mathcal{S}_{u(U)}$ such that $\beta = \beta' \circ \iota$, which implies that $\beta$ is strongly cartesian in $\mathcal{S}$ by Categories, Lemma 02XL.

Finally, we have to show that given $(U', y')$ and $U \to U'$ we can find a strongly cartesian morphism $(U, y) \to (U', y')$ in $u^p\mathcal{S}$ lifting the morphism $U \to U'$. This follows from the above as by assumption we can find a strongly cartesian morphism $y \to y'$ lifting the morphism $u(U) \to u(U')$. $\square$